

15	C	$\text{Max } a = (2\pi \times 120)^2 \times 40 \times 10^{-3} = 22716 \text{ ms}^{-2}$ and $\text{max } v = (2\pi \times 120) \times 40 \times 10^{-3} = 30.1 \text{ ms}^{-1}$
16	D	Q leads P by $\frac{\pi}{2}$ rad. Equivalently, P leads Q by $\frac{3\pi}{2}$ rad.
17	A	$\theta_{\min} \approx \frac{\lambda}{r} = \frac{550 \times 10^{-9}}{35 \times 10^{-2}} = 1.57 \times 10^{-6} \text{ rad}$